Week 7 Progress Report

BSE4104 Emerging Trends in Software Engineering

AI-Native & Agentic Engineering Capstone Project
SME Procurement-Preparation Agent

GROUP J EVENING

1. Work Completed Against Weekly Objectives

    - Defined clear evaluation criteria: Task Completion, Groundedness, Tool Selection, Instruction Following, Safety and Latency.
    - Created a 30-scenario evaluation dataset covering Normal, Edge, Missing/Incorrect Information, Adversarial, Tool Failure, Unauthorized/Safety, RAG/Policy and Combined cases.
    - Strengthened observability by ensuring every agent run produces a structured trace saved under evidence/traces/.
    - Verified and reinforced key guardrails:
        - Input/output validation
        - Tool allow-list enforcement
        - Agent iteration limits (max 8 turns)
        - Human approval gate (no autonomous approval or ordering)
        - Basic prompt-injection resistance
    - Executed the full evaluation set and recorded results.

- Produced a Failure Catalogue containing five genuine failures, each with root cause analysis, remediation, and successful re-test evidence.
- Committed the evaluation dataset, results, Failure Catalogue, and traces to GitHub.

2. Key engineering decisions

Decision	Rationale
30 diverse scenarios with explicit categories	Meets the course minimum and provides balanced coverage of normal, edge, adversarial and safety cases.
Structured JSON traces for every run	Enables precise debugging, supports the Failure Catalogue and satisfies observability requirements.
Failure Catalogue focused on genuine observed issues	Demonstrates honest evaluation and continuous improvement rather than hiding problems.
Hard enforcement of tool allow-list and iteration limits	Provides technical (not just prompt-based) guarantees that the agent remains bounded.
Re-testing after every remediation	Ensures that fixes actually work and that regressions are caught early.

3. Challenges encountered and response

Challenge	Response
Model occasionally hallucinated data after a tool error	data after a tool errorStrengthened system prompt and added clearer error-handling instructions. Re-tested successfully.
Model attempted to call non-existent tools under adversarial prompting	Reinforced allow-list instructions and improved orchestrator feedback.
Preferred-supplier ranking not applied in one test run	Fixed import/integration issue in get_quotations. Re-tested successfully
Agent did not always stop cleanly at max iterations	Updated prompt + orchestrator to force a clear hand-over summary.
Keeping evaluation consistent across team members	Used a shared scenario CSV and a standard results template.

4. Repository and task management evidence
    - GitHub Repository: https://github.com/Nicholas-Muwonge/SME-Procurement-Agent
    - Key additions this week:
        - docs/evaluation/Week7_30_Scenario_Evaluation_Dataset.csv
        - docs/evaluation/Week7_Failure_Catalogue.md
        - Updated system prompt and minor guardrail improvements in src/orchestrator.py
        - Evaluation results summary
    - ClickUp Board: https://app.clickup.com/1200420000004171/v/b/6-1200420000007249-2

5. Plan for Week 8
   Focus: Hardening, Final Release and Demonstration

    - Review all requirements, architecture, prompts, RAG, tools agent limits, memory, tests, security and documentation.
    - Fix any remaining critical defects and document known limitations.
    - Prepare a tagged, reproducibly runnable release with clear README and run instructions.
    - Complete the final engineering report
    - Prepare the presentation deck and live demonstration.
    - Finalise ClickUp board and repository evidence pack.